STaR-Body: Identity-Preserving Evidence Routing for Spatiotemporal Human Mesh Refinement

Supplementary Material

Anonymous CVPR submission

This supplement documents the implementation (Sec. S1), the objective and schedule (Sec. S2), the evaluation protocol and its deviations from detector-based pipelines (Sec. S3), the statistical procedure (Sec. S4), the observation-head audit and the archived records (Sec. S5), full tables over seeds and splits (Sec. S6) and per-sequence results (Sec. S7). As in the main paper, every number of the new experiments is inserted by `scripts/analyze_results_c11.py`; a red **TBD** marks a value that has not been produced yet.

S1. Implementation details

Token interfaces. Table S1 lists the interfaces of the model. Each target window contains eight frames at positions 4–11 of a 17-frame video context. HMR 2.0a supplies a 192-token image grid per frame; each of the five VACE latent slices is pooled from a 6×6 grid to 2×3 cells, giving 30 memory tokens. The dual-order decoder has width 256 and eight heads, with four blocks in each of three weight-shared iterations. In the refiner, local image samples are projected to 64 channels and combined with the 256-dimensional pose query and the 22-dimensional body state (9 rotation, 10 shape, 3 camera values per joint) into a 128-dimensional joint feature.

Representation	Tokens	Width	Note
HMR 2.0a image grid	$T \times 192$	1,280	frozen
VACE latent grid	5×36	1,536	frozen, cached
Pooled video memory	30	256	
Body queries	$T \times 26$	256	24 pose, shape, camera
Evidence tokens (one source)	$T \times 17$	$7 \to 128$	main Eq. (2)
Evidence tokens (head + external, B6)	$T \times 34$	$7 \to 128$	separate identity embeddings
Joint feedback	$T \times 24$	128	zero at initialisation

Table S1. Token interfaces; $T = 8$ target frames. Image and video feature extraction is frozen.

Router tensors. For a batch of B windows, a source is given by residuals and projections $[B, T, 17, 2]$, the log-confidence feature and the confidence $[B, T, 17, 1]$, frame validity $[B, T]$ and slot validity $[B, T, 17]$. For external keypoints (u, v, c), the log-confidence feature is $-2 \log \max(c, 0.0025)$ clipped to $[-6, 6]$, the confidence feature is c, and a slot is valid when it is finite and c exceeds a minimum confidence (zero in all reported runs); frames with a non-positive or non-finite camera scale are invalid. The token MLP is $\mathrm{Linear}(7, 128)$–SiLU–$\mathrm{Linear}(128, 128)$, the encoded tokens have shape $[BT, 17, 128]$, the 24 joint queries $[BT, 24, 128]$, and the additive attention mask $[4BT, 24, 17]$ holds the bias $-D(j, a(l))/2$ and $-\infty$ for invalid slots. If no slot of a frame is valid, slot 1 is temporarily unmasked to keep the softmax finite and the output of that frame is set to zero. The output $\mathrm{Linear}(128, 128)$ without bias is zero-initialised and yields $[B, T, 24, 128]$, added to the joint features at every refinement iteration. The anchors $a(l)$ map the five COCO facial landmarks to the SMPL head joint and shoulders, elbows, wrists, hips, knees and ankles to the corresponding SMPL joints; D is the shortest-path distance on the 24-joint SMPL tree. Variant B5 permutes the 17 bias columns with the fixed derangement $(9, 13, 5, 16, 11, 2, 15, 0, 12, 4, 14, 1, 8, 6, 3, 10, 7)$ (zero-based), which has no fixed point. The router of each run is initialised on the CPU from its own generator with seed $370026 + (s - 2026)$ for training seed s, so that it does not change the random stream of the rest of the model; for $s = 2026$ this is the initialisation of the archived C10 router.

Parameter counts. Table S2 counts the parameters of the feedback module that receive gradient, obtained by instantiating every configuration of `models/evidence_router_v43.py` with width 128 and back-propagating through a forward pass. For checkpoint compatibility with C10, the module always instantiates the full C10 parameter set; tensors that a configuration does not use receive no gradient and are never updated ("instantiated" column).

Variant	Components that receive gradient	Used	Instantiated
A0 none	–	0	–
A1 mean (C00)	Linear$(4, 128)$ on averaged residual, log-variance, visibility	512	512
A2 head, routed (C10)	tokens 17,536; identity 2,176; schema 256; queries 3,072; attention 66,048; output 16,384	105,472	105,472
B4, B3, B5, C4 routed	as A2, with the external identity embedding	105,472	107,648
B2 set	tokens, queries, attention, output (no identity, no schema)	103,040	107,648
B6 head + external	as A2 plus external identity 2,176	107,648	107,648
B1, C1 pooled	tokens 17,536; LayerNorm–128–256–128 MLP 66,176; output 16,384	100,096	173,824
B7 ordered vector	tokens 17,536; Linear$(17{\cdot}128, 128)$–SiLU–LayerNorm–Linear 295,424; output 16,384	329,344	403,072

Table S2. Feedback-module parameters (width 128). Attention: four heads, input and output projections with biases.

External keypoints. ViTPose++-H (COCO-17 expert head, transformers implementation) runs top-down on the central 192×256 region of the same 256×256 crop that the HMR 2.0a backbone sees; coordinates are normalised as $x/256 - 0.5$, the convention of the body projection. The keypoints are cached once per split and never fine-tuned. For C1 and C4, the OpenPose-18 detections distributed with 3DPW (the `poses2d` field) are mapped to COCO-17 with their confidences; these are the same detections that define the crop box. No ground-truth 2D or 3D joint enters either cache.

Bounded rotation updates. Write $w = \tanh u \in (-1, 1)^3$. Then $\|w\| < \sqrt{3}$ and, since $s \mapsto s/\sqrt{1+s^2}$ is increasing, $\|\omega\| = \alpha \|w\|/\sqrt{1+\|w\|^2} < \alpha\sqrt{3}/2$. With $\alpha = 2^\circ$ (root) and 5° (other joints) each step is below 1.73° and 4.33°, and by the triangle inequality for the geodesic distance on $SO(3)$ three steps move a joint by less than 5.2° and 13.0° relative to the refiner input. This bounds the refinement displacement, not the error with respect to the ground truth. The archived manuscript stated the looser bounds 6° and 15°, which also hold.

S2. Objective and optimisation

Term	Weight
Joint / vertex error	1 / 1
Rotation / shape error	0.05 / 0.001
Velocity error	0.5
Within-window / boundary acceleration	3.012855 / 0.987145
Procrustes-aligned joint auxiliary	0.2
Intermediate base geometry	0.25
Shape consistency	0.01
Update regularisation	0.001
2D reprojection, observation and visibility terms	0

Table S3. Loss coefficients, identical for all variants. Coordinate losses are in metres; the two acceleration coefficients implement the count-weighted within/boundary partition and sum to four.

Table S3 lists the loss coefficients. Joint and vertex terms use Euclidean errors; rotation and shape supervision use squared errors. Velocity and acceleration act on the hip-centred prediction-minus-target joint error ϵ_t; at an annotation-adjacent triplet the acceleration error is $\frac{1}{14}\sum_{j=1}^{14}\|\epsilon_{t-1,j} - 2\epsilon_{t,j} + \epsilon_{t+1,j}\|_2$. Every person-frame is owned by one window; triplets inside an owner window and triplets across owner windows form disjoint sets, and the boundary set is traversed cyclically with its owner contexts forwarded separately, so that the final-output motion objective covers complete sequences. All 2D terms have weight zero; this is why the inherited observation head never received gradient (Sec. S5).

Every variant starts from the same parent and runs 1,000 optimiser steps with AdamW (weight decay 10^{-4} on matrices), gradient-norm clipping at one and an effective batch of 16 windows (batch one, 16 accumulation steps), which amounts to 16,000 training-window draws and 16,000 boundary-triplet draws, covering all 2,909 training windows and all 5,548 boundary triplets. Neural modules run under BF16 autocast and geometry in FP32. All parameters outside the decoder backbone (track-shape residual, recurrent refiner with its part/time blocks, and the feedback module) train at 5×10^{-5} from step zero; at step 300 the body, root and shape heads and the output normalisation become trainable at 5×10^{-6}, and at step 800 the last two decoder blocks, the image attention and the inherited feedback projection at 10^{-6}. The reference, feature encoders, camera head and observation head stay frozen. Learning rates use 200 warm-up steps and a cosine schedule with a 6,000-step horizon and a 10% floor; at update k the EMA decay is $\min(0.999, (1+k)/(10+k))$. The evaluated weights are the EMA at step 1,000 for every variant and seed. **[TODO: Add GPU type, wall-clock time per run and peak memory from the run records (`run_config.json`, evaluation `resources`).]**

S3. Evaluation protocol and deviations

Metrics. Predicted bodies use the neutral SMPL model and targets the gendered SMPL model. Joints are the 14 LSP joints obtained from mesh vertices with the H36M regressor. `mpjpe` subtracts the H36M pelvis (joint 0) from both skeletons, as in SPIN, CameraHMR and ReFit; `mpjpehip` subtracts the midpoint of the two LSP hips, as in the VIBE evaluator, WHAM and GVHMR. `pampjpe` applies a per-frame similarity Procrustes alignment. `pve` is our historical vertex error without any alignment, `pvepelvis` subtracts each body's H36M pelvis from its vertices (CameraHMR, ReFit, PromptHMR), and `pvehip` subtracts each body's LSP hip midpoint (WHAM, GVHMR). VIBE, TCMR and GLoT compute PVE against neutral-gender ground-truth bodies without alignment, which we do not reproduce. `accel` is the mean over all annotation-adjacent triplets (including triplets across inference windows) of the second difference of the hip-centred prediction-minus-target error, in mm/frame^2; `accelms` $= 0.9\times$`accel` is the same quantity in m/s^2 at 30 fps. Aggregates weight person-frames (or triplets) equally; sequence errors are summed and divided by the total count. Inference and SMPL computation run in FP32 with double-precision sums, without flip averaging or smoothing.

Deviations from detector-and-tracker pipelines. (i) Person crops are derived from the 2D person annotations of 3DPW with a 25% margin rather than from a person detector. (ii) Track identities are the dataset's person identities; the track prototype used by the shape residual is computed offline from the whole track and is therefore non-causal. (iii) Frame validity masks follow the dataset's camera-pose validity flags. (iv) Earlier models of this project were evaluated on the 3DPW test split before this study (we cannot undo that exposure); in this study, test evaluation is gated by validation locks and no decision used test results. (v) The historical unaligned PVE of earlier drafts is not comparable to recent literature; we report it only for continuity. (vi) The archived C10 numbers were selected on validation, the split they are reported on. These deviations apply equally to all variants of the controlled study.

S4. Statistical procedure

For variant V, control C, metric m, paired seeds $s \in \mathcal{S}$ and sequences $q \in \mathcal{Q}$ with counts n_q (person-frames, or triplets for acceleration), let $E^V_{s,q}$ be the summed error of sequence q. The reported difference is

$$\Delta = \frac{\frac{1}{|\mathcal{S}|}\sum_s \sum_q \left(E^V_{s,q} - E^C_{s,q}\right)}{\sum_q n_q}, \tag{S1}$$

i.e. the frame-weighted aggregate difference averaged over seeds. For the 95% interval we draw $|\mathcal{S}|$ seeds with replacement and then $|\mathcal{Q}|$ sequences with replacement, recompute Δ on the resample (with the counts of the drawn sequences in the denominator), repeat 20,000 times and take the 2.5% and 97.5% quantiles (percentile bootstrap [1], random seed 20261101). Resampling sequences rather than frames respects the strong correlation of frames within a sequence; resampling seeds propagates training variance. With three seeds the seed level of the bootstrap is coarse, so intervals may be narrower than the true uncertainty about the training distribution. "Wins" counts sequences whose seed-averaged difference is negative. The paired t-test uses the per-seed aggregate differences ($n = 3$, two degrees of freedom). We call a difference reliable only if its interval excludes zero and all three seeds agree in sign; this rule and the list of comparisons were fixed before test evaluation. We do not correct for the 14 comparisons; the two primary ones are B4 *vs.* B1 and B4 *vs.* B7.

S5. Observation-head audit and archived records

Audit. The parent's observation head predicts landmark coordinates as a correction to the projection of the frozen reference body, a log-variance and a visibility logit. It is created with zero coordinate and log-variance weights, log-variance bias -4 and zero visibility weights. In every training stage that produced the parent, C00 and C10, all 2D losses have weight zero (Tab. S3), and the stage contracts forbid enabling them; the head therefore receives no gradient. If the head is still at its initialisation, its coordinates equal the reference projection, its log-variance is -4 and its visibility is $\sigma(0) = 0.5$ for every input, and the residual routed by C10 is $p(x^{\text{ref}}) - p(x^r)$. `scripts/inspect_checkpoint_v43.py` checks these conditions on the stored weights and EMA weights of each checkpoint. **[TODO: Insert the verdict for the archived parent, C00 and C10 checkpoints (expected: "UNTRAINED" for both weight fields), the maximum absolute coordinate weight and the log-variance bias. If any head turns out to be trained, rewrite Sec. 3.2 of the main paper accordingly.]**

Archived records. Table S4 reproduces the archived single-seed validation records (10,413 person-frames, 10,226 triplets; validation was also the selection split). C00 values are transcriptions of contemporaneous run records whose raw evaluation files are not archived. At 1,000 steps, the C10 student is worse than the parent in MPJPE and its gain appears only in the EMA weights. From the archived per-sequence error sums of the parent and C10, a sequence-level bootstrap (12 sequences, 20,000 resamples, single seed) gives C10 minus parent differences of -0.07 mm MPJPE (95% interval $[-0.21, +0.03]$; lower on 9 of 12 sequences), -0.03 mm PA-MPJPE ($[-0.12, +0.05]$; 10 of 12) and -0.21 mm unaligned PVE ($[-0.32, -0.11]$; 11 of 12). These differences include the effect of 1,000 additional steps with EMA, which A0 isolates in the new study. The C10-versus-C00 difference cannot be bootstrapped because C00 sequence records are not archived.

Model	Weights	MPJPE	PA-MPJPE	PVE (unaligned)	Accel
Spatiotemporal reference	selected	61.42	37.78	73.29	21.51
Parent	–	61.26	37.64	73.02	21.48
C00 mean feedback, 1,000 steps*	student	61.39	37.65	73.20	21.44
C00 mean feedback, 1,000 steps*	EMA	61.26	37.64	73.05	21.45
C10 joint feedback, 1,000 steps	student	61.42	37.65	73.03	21.43
C10 joint feedback, 1,000 steps	EMA	61.19	37.61	72.82	21.44
Re-evaluation with the v43 code path					
Parent P	–	**TBD**	**TBD**	**TBD**	**TBD**
C00	archived	**TBD**	**TBD**	**TBD**	**TBD**
C10	archived	**TBD**	**TBD**	**TBD**	**TBD**

Table S4. Archived single-seed 3DPW validation records (mm; Accel in mm/frame^2) and their re-evaluation through the code used for the new study, which should reproduce them. *Transcribed from run records. **[TODO: State whether the re-evaluation reproduces the archived values and which weight field each archived checkpoint stores.]**

S6. Full results over seeds

Tables S5 and S6 give every variant under every metric (mean$\pm$std over seeds; reference checkpoints are single models). Tables S7 and S8 give the paired comparisons with bootstrap intervals and paired t-test p-values.

S7. Per-sequence results

Tables S9 and S10 list the seed-averaged per-sequence differences of the main method to the pooled control, the ordered-vector control and no feedback (mm; negative favours B4).

References

[1] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993. 3



Figure S1. Paired differences (first minus second variant; MPJPE and PA-MPJPE, mm) for all pre-registered comparisons on validation (circles) and test (squares), with 95% hierarchical cluster-bootstrap intervals over seeds and sequences. Negative values favour the first variant.

Variant	n	MPJPE	$\text{MPJPE}_{\text{hip}}$	PA-MPJPE	PVE	$\text{PVE}_{\text{pelvis}}$	PVE_{hip}	Accel (mm/f^2)	Accel (m/s^2)
P parent	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C00 archived	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C10 archived	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
A0 none	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
A1 mean	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
A2 head routed	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B1 pooled	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B7 ordered	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B4 routed	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B2 set	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B3 no bias	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B5 deranged	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B6 head+ext.	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C4 OP routed	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C1 OP pooled	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table S5. All variants on 3DPW **validation** (12 sequences, 10,413 person-frames, 10,226 triplets); mm unless noted.

Variant	n	MPJPE	$\text{MPJPE}_{\text{hip}}$	PA-MPJPE	PVE	$\text{PVE}_{\text{pelvis}}$	PVE_{hip}	Accel (mm/f^2)	Accel (m/s^2)
P parent	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C00 archived	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C10 archived	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
A0 none	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
A1 mean	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
A2 head routed	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B1 pooled	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B7 ordered	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B4 routed	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B2 set	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B3 no bias	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B5 deranged	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
B6 head+ext.	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C4 OP routed	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
C1 OP pooled	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table S6. All variants on 3DPW **test** (24 sequences, 35,515 person-frames, 34,752 triplets); mm unless noted.

Comparison	ΔMPJPE [95% CI]	p	ΔPA-MPJPE [95% CI]	$\Delta\text{PVE}_{\text{pelvis}}$ [95% CI]	p	ΔAccel [95% CI]	Wins (PVE)
B4 *vs.* B1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B7	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B2	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B3	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B5	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* A2	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B6 *vs.* B4	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B1 *vs.* A1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
A2 *vs.* A1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
A1 *vs.* A0	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
A2 *vs.* A0	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* A0	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
C4 *vs.* C1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* C4	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD

Table S7. Paired differences on **validation** (first minus second; mm, Accel in mm/frame^2; Wins out of 12 sequences). Intervals: hierarchical cluster bootstrap over seeds and sequences; p: paired t-test over three seeds.

Comparison	ΔMPJPE [95% CI]	p	ΔPA-MPJPE [95% CI]	$\Delta\text{PVE}_{\text{pelvis}}$ [95% CI]	p	ΔAccel [95% CI]	Wins (PVE)
B4 *vs.* B1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B7	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B2	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B3	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* B5	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* A2	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B6 *vs.* B4	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B1 *vs.* A1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
A2 *vs.* A1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
A1 *vs.* A0	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
A2 *vs.* A0	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* A0	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
C4 *vs.* C1	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD
B4 *vs.* C4	TBD TBD	TBD	TBD TBD	TBD TBD	TBD	TBD TBD	TBD

Table S8. Paired differences on **test** with p-values and PA-MPJPE/Accel intervals (complements Tab. 3 of the main paper; Wins out of 24 sequences).

	B4 *vs.* B1		B4 *vs.* B7		B4 *vs.* A0	
Sequence	ΔMPJPE	$\Delta\text{PVE}_{\text{pelvis}}$	ΔMPJPE	$\Delta\text{PVE}_{\text{pelvis}}$	ΔMPJPE	$\Delta\text{PVE}_{\text{pelvis}}$
downtown_arguing_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_bar_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_bus_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_cafe_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_car_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_crossStreets_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_downstairs_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_enterShop_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_rampAndStairs_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_runForBus_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_runForBus_01	TBD	TBD	TBD	TBD	TBD	TBD
downtown_sitOnStairs_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_stairs_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_upstairs_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_walkBridge_01	TBD	TBD	TBD	TBD	TBD	TBD
downtown_walkUphill_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_walking_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_warmWelcome_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_weeklyMarket_00	TBD	TBD	TBD	TBD	TBD	TBD
downtown_windowShopping_00	TBD	TBD	TBD	TBD	TBD	TBD
flat_guitar_01	TBD	TBD	TBD	TBD	TBD	TBD
flat_packBags_00	TBD	TBD	TBD	TBD	TBD	TBD
office_phoneCall_00	TBD	TBD	TBD	TBD	TBD	TBD
outdoors_fencing_01	TBD	TBD	TBD	TBD	TBD	TBD
All (frame-weighted)	TBD	TBD	TBD	TBD	TBD	TBD

Table S9. Per-sequence differences on 3DPW **test** (seed-averaged; mm).

	B4 *vs.* B1		B4 *vs.* B7		B4 *vs.* A0	
Sequence	ΔMPJPE	$\Delta\text{PVE}_{\text{pelvis}}$	ΔMPJPE	$\Delta\text{PVE}_{\text{pelvis}}$	ΔMPJPE	$\Delta\text{PVE}_{\text{pelvis}}$
courtyard_basketball_01	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
courtyard_dancing_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
courtyard_drinking_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
courtyard_hug_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
courtyard_jumpBench_01	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
courtyard_rangeOfMotions_01	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
downtown_walkDownhill_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
outdoors_crosscountry_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
outdoors_freestyle_01	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
outdoors_golf_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
outdoors_parcours_00	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
outdoors_parcours_01	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
All (frame-weighted)	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

Table S10. Per-sequence differences on 3DPW **validation** (seed-averaged; mm).